

In this short document we try to give a self-contained introduction to the problem. We are interested in the asymptotics of the following integral

$$\frac{1}{n} \log \int_{[0,\infty)^n} \exp(-x^T M_n x / 2) dx. \quad (1)$$

Here, M_n is an n -by- n symmetric, positive-definite matrix with equal diagonal entries. Moreover, in the expansion of quadratic form $x^T M_n x$, each x_i appears exactly same number of times in cross terms besides diagonal terms for all n . For example, if $n = 4$, then $x_1 x_2 + x_3 x_4$ and $x_1 x_3 + x_2 x_4$ are quadratic forms that satisfy the condition.

M_n has some additional properties that might aid us in the computation. We will then give a brief description of the overall scheme of the problem, why do we care about this integral, and what additional properties does the matrix M_n have.

The original problem starts with the familiar SK model $H : \{-1, +1\}^N \rightarrow \mathbb{R}$:

$$H(\sigma) := \sum_{1 \leq i < j \leq N} \sigma_i \sigma_j W_{ij}, \quad (2)$$

where $W = (W_{i,j})_{N \times N}$ is a symmetric matrix with zero diagonal such that $(W_{i,j})_{1 \leq i < j \leq N}$ are independent standard normal Gaussians. (We omit the normalization factor because it won't matter.) The energy landscape problem of spin glass models has been a vital topic of probability and statistics. In our case, we are particularly interested in the “potential energy well” of the model, a.k.a. the local optimum. For a configuration σ , we are at a local optimum if flipping any coordinates will cause the energy H to increase. Intuitive speaking, the energy increases whenever we take a single step away from the original configuration σ . The “potential energy well” has radius one.

Clearly, the difference between the original energy and the energy after taking 1 step is a Gaussian random variable. For any N , there are N possible steps we can take. The configuration is a local optimum if and only all the differences are greater than zero. This gives us N Gaussian variables. Turns out that the covariance matrix depends only on N and its inverse can be written as an expression of $\|x\|_1$ and $\|x\|_2$. The probability of the 1-step local optimum's can and has already been computed. This is the integral in (1). M is the inverse of the covariance matrix. The integral is missing the determinant of the covariance matrix in its denominator, but that has been taken care of.

To generalize the result, we want to compute the radius m local optimum for any fixed positive integer m . Instead of taking one step, now we are allowed to take m steps. We get a $\binom{N}{m}$ -by- $\binom{N}{m}$ covariance matrix and that makes the computation of (1) much more difficult (we have $n = \binom{N}{m}$). The good news is that we can compute the inverse of the covariance matrix, a.k.a. the matrix M_n . Since M_n has dimension $\binom{N}{m}$ -by- $\binom{N}{m}$, which corresponds to the m coordinates we flip (or the m steps we take), its row and column can be indexed by subsets of $[N] := \{1, 2, \dots, N\}$ with cardinality m . (Such indexing also has combinatorial meaning by our discussion.)

We have already shown that both the covariance matrix C_n and its inverse M_n (the matrix in the integral) satisfy the following:

Let S and T be two given m -subsets of $[N]$, $|S \cap T| = l$. The entry at row S and column T only depend on m (given and fixed), N , and l .

Since $0 \leq l \leq m$, M_n only has at most $m + 1$ distinct values for its entries. In its quadratic form, every x_i participates equally. We try to make a breakthrough by computing the case where $m = 2$. Let $\{a, b\}$ and $\{c, d\}$ be two 2-subsets of $[N]$. $\{a, b\} \cap \{c, d\} = l$ and l can only be 0, 1 or 2. ($l = 2$ corresponds to the diagonal terms.) $n = \binom{N}{2}$ and for N very large, the matrix M_n asymptotically satisfies the following:

1. For N very large, M_n 's diagonal terms are asymptotically equal to $\frac{1}{4}$.
2. If $\{a, b\} \cap \{c, d\} = 1$, the entry at row $\{a, b\}$ and column $\{c, d\}$ is equal to $-\frac{1}{4N}$.
3. If $\{a, b\} \cap \{c, d\} = 0$, the entry at row $\{a, b\}$ and column $\{c, d\}$ is equal to $\frac{1}{2N^2}$.

(1) can thus be rewritten as:

$$\binom{N}{2}^{-1} \log \int_{[0, \infty)^{\binom{N}{2}}} \exp(-x^T M_n x / 2) dx. \quad (3)$$

It would be of great help that we can get a large deviation principle established for the exponential quadratic form. So far we are not sure what kind tools can be used to compute such integral. The quadratic form is not the sum of i.i.d. random variables and not a stationary process. Most tools we have for large deviations are not applicable here, making this integral the biggest roadblock of the problem.